Quiz 8

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no devices

Name: _____ UT EID: _____

Four data points in $\mathbb{R}^2$: $x_1 = (3, 3)$, $x_2 = (-3, -3)$, $x_3 = (1, -1)$, $x_4 = (-1, 1)$. Their mean is 0. Find a unit vector $q \in \mathbb{R}^2$ and numbers c_1, c_2, c_3, c_4 that minimize

$$\frac{1}{4} \sum_{i=1}^{4} \|x_i - c_i q\|^2.$$

Show your work. [10 pts]

Answer:

Total: 10 points. Lowest two quiz scores are dropped at the end of the semester.